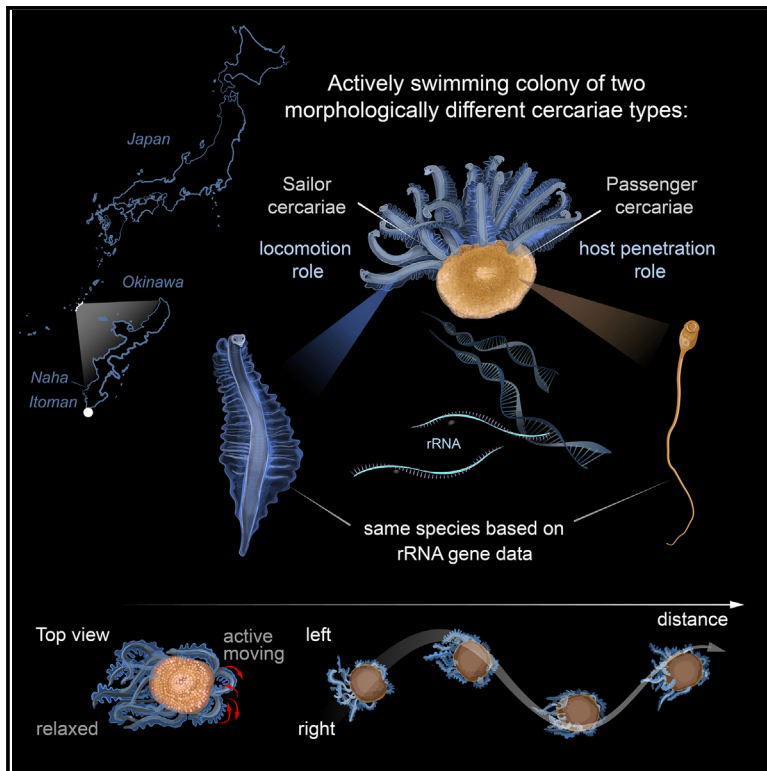


Current Biology

Polymorphic parasitic larvae cooperate to build swimming colonies luring hosts

Graphical abstract



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In brief

Digenea flatworms have free-living larvae, cercariae, which exhibit diverse structures and behaviors. Krupenko et al. discover a swimming colony consisting of two types of cercariae: “sailors” for mobility and smaller “passengers” for infection. Genetic analysis reveals that both types belong to the same species in the Acanthocolpidae family.

Highlights

- An enigmatic aggregate of worm-looking animals was collected near Okinawa shores
- The colony consisted of two different types of digenean cercariae larvae
- The locomoting sailor and passive passenger cercariae belong to the same species
- This represents the first case of labor division in digenean larvae



Report

Polymorphic parasitic larvae cooperate to build swimming colonies luring hosts

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SUMMARY

Parasites have evolved a variety of astonishing strategies to survive within their hosts, yet the most challenging event in their personal chronicles is the passage from one host to another. It becomes even more complex when a parasite needs to pass through the external environment. Therefore, the free-living stages of parasites present a wide range of adaptations for transmission. Parasitic flatworms from the group Digenea (flukes) have free-living larvae, cercariae, which are remarkably diverse in structure and behavior.^{1,2} One of the cercariae transmission strategies is to attain a prey-like appearance for the host.³ This can be done through the formation of a swimming aggregate of several cercariae adjoined together by their tails.⁴ Through the use of live observations and light, electron, and confocal microscopy, we described such a supposedly prey-mimetic colony comprising cercariae of two distinct morphotypes. They are functionally specialized: larger morphotype (sailors) enable motility, and smaller morphotype (passengers) presumably facilitate infection. The analysis of local read alignments between the two samples reveals that both cercaria types have identical 18S, 28S, and 5.8S rRNA genes. Further phylogenetic analysis of these ribosomal sequences indicates that our specimen belongs to the digenean family Acanthocolpidae, likely genus *Pleorchis*. This discovery provides a unique example and a novel insight into how morphologically and functionally heterogeneous individuals of the same species cooperate to build colonial organisms for the purpose of infection. This strategy bears resemblance to the cooperating castes of the same species found among insects.⁵

RESULTS

In 2018 (11th of April), Japanese underwater photographer Ryo Minemizu spotted an actively swimming 5-mm-large animal at 16 m depth off the shore of Kiyan Cape (Okinawa, Japan) (Figure 1). After posting the images of this animal to social networks, hundreds of professional biologists failed at identification to the level of a phylum.

2 years later, we assembled an international team to take on this identification challenge, as Ryo collected a single individual sample of the animal and stored it in a 10% formaldehyde solution at room temperature for about 2 years. Despite significant technical challenges associated with small size and strong fixation, we successfully performed morphological and genetic studies, which revealed that we are dealing with a previously

unknown aggregate of polymorphic digenean cercaria larvae (Trematoda: Digenea).

After microscopy-assisted exploration of the unstained specimen, we established that the aggregate is composed of two types of cercariae, which are henceforth referred to as the “sailors” (large ones) and the “passengers” (small ones) (Figures 1C–1H). The single aggregation we studied included about 20 sailors and over 1,000 passengers. Passenger cercariae were arranged in a hemisphere, with their tails forming the bulk and their bodies at the periphery. The tail tips of the sailor cercariae were anchored at the flattened side of the hemisphere, resembling snake hair on the Gorgon’s head. The passengers were much smaller volume-wise, as compared to large sailors. The aggregate resembles a colony, being finely structured and optimized for functional performance at different levels.



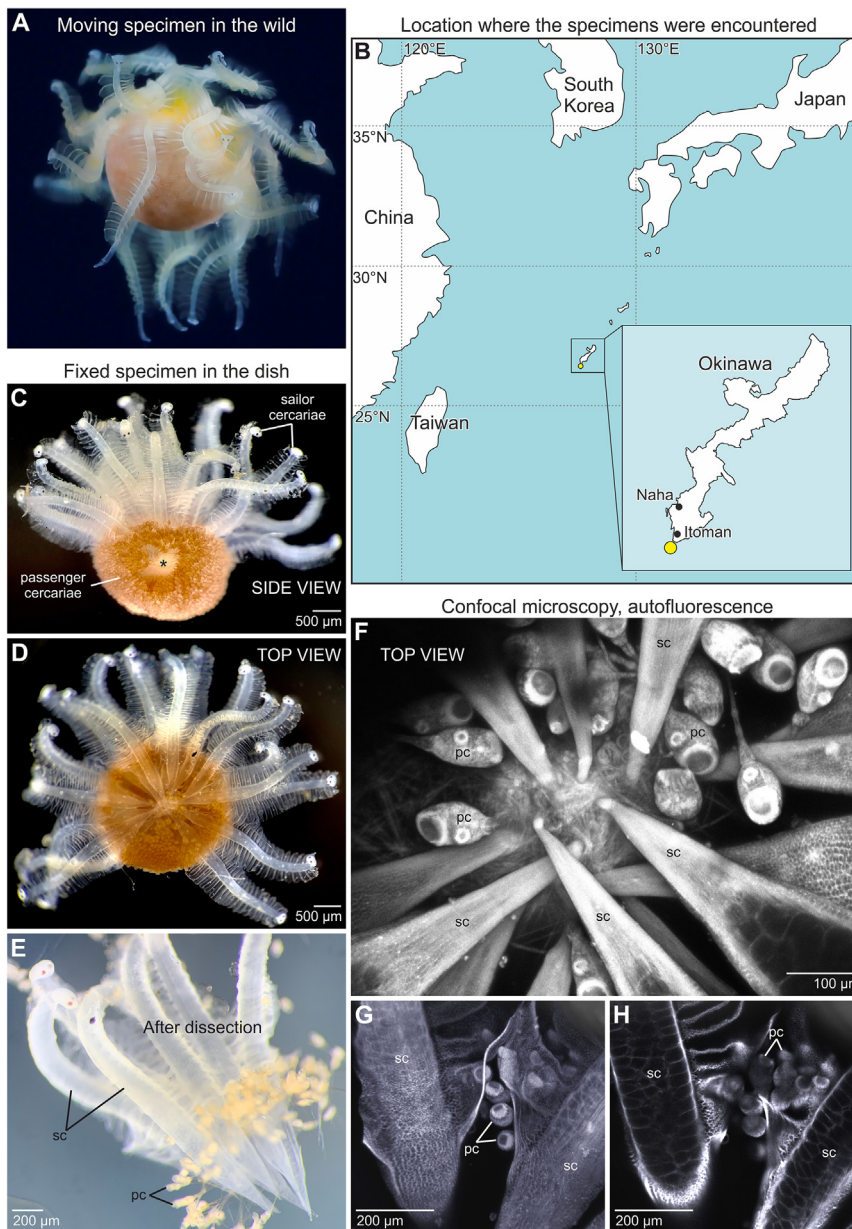


Figure 1. General structure of aggregate

(A) Moving specimen in the wild.
(B) Sampling location.
(C) Fixed specimen in the dish, side view; hemisphere of passenger cercariae (pc) partially disrupted (asterisk).
(D) Same as (C), top view.
(E) Dissected fixed specimen.
(F–H) Confocal images of tail tips of sailor cercariae (sc) embedded into bulk of hemisphere made of passenger cercariae tails, autofluorescence.

colony with integrated structure and coordinated behavior. We should note that studying a single specimen does not allow us to estimate the variations in locomotion patterns or in proportion between sailor and passenger cercariae.

Both passengers and sailors are readily distinguishable as the digenean cercaria larvae by the presence of tail, oral, and ventral suckers. However, they differ drastically in many morphological characteristics (Figure 3; for full formal description see Table S1). The passengers' body is stumpy with collar-like thickening in the anterior part, while the sailors' body is dorsoventrally flattened. Tegument of both cercaria types bears spines, though the spination pattern slightly differs in sailors and passengers. Voluminous cerebral ganglia are present in both passengers and sailors, but only the latter possess a pair of large, pigmented eyespots. Each eyespot has two pigment-cup cells and two lenses (Figure 3L). For the passengers, we partially visualized nervous system by antibodies for glutamic acid decarboxylase, which labels GABAergic neurons and their projections, and found it well developed in both body and tail (Document S1). Of the digestive system primordium, only the pharynx is

recognizable in both types of cercariae. The excretory vesicle is triangular in the passengers and oval in sailors. Genital primordium is present in both types of cercariae; it is compact, triangular in the passengers, while in sailors it has a complex shape (Figures 3F and 3M). Only the passengers have penetration glands. They are six pairs, large, located in the forebody, and without nuclei; each gland has its own duct (Figures 3F, 3G, and 3I). The tails of the passenger cercariae are simple and thin, with maximal width at the base. Meanwhile the tails of the sailors are enormous, uniformly wide except for the hind part. They have strong longitudinal musculature and wide lateral fins forming undulatory plications, extending from the tail base to the posterior $\frac{1}{6}$ of the tail's length.

Although the true coloration of aggregate was lost due to fixation, from the live photos and videos it is evident that every sailor

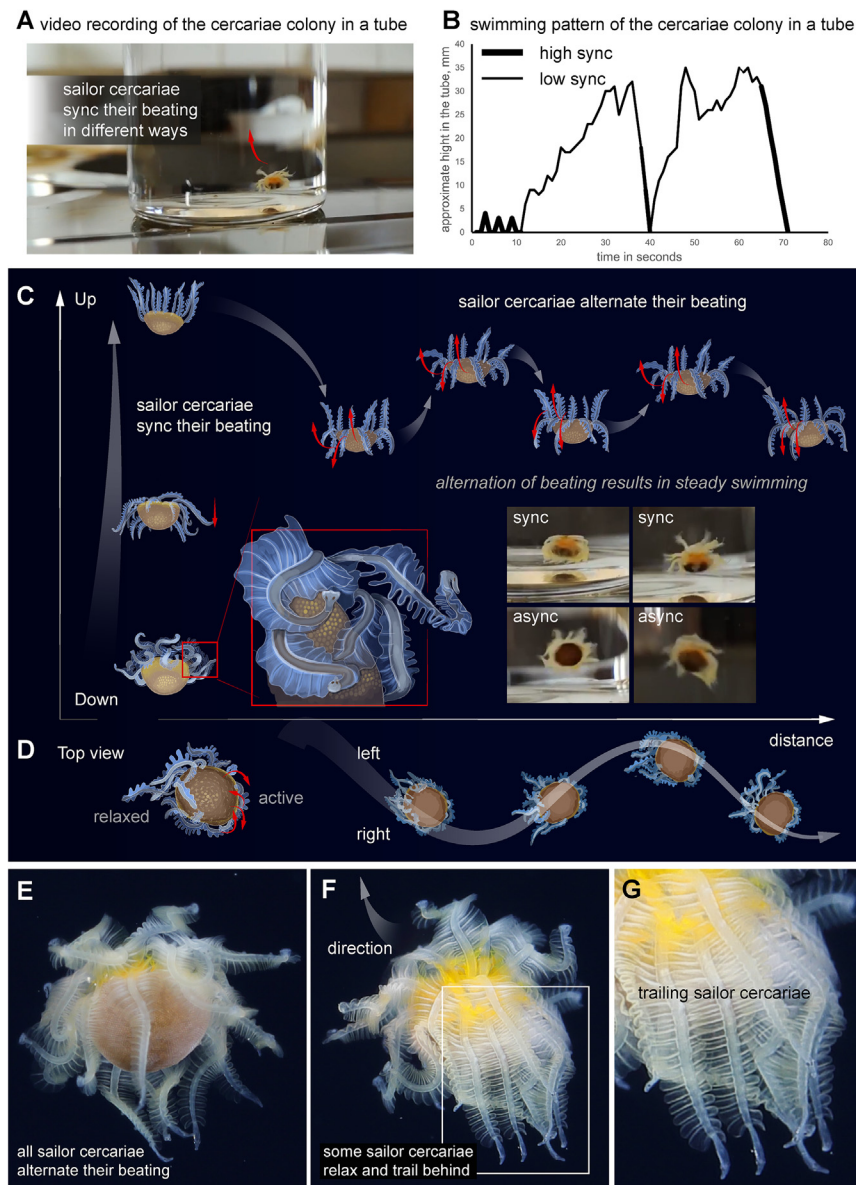


Figure 2. Locomotion of cercariae aggregate

(A) Live specimen in glass tube.
(B) Swimming pattern of aggregate in tube.
(C and D) Schematic representation of aggregate locomotion.
(E–G) Synchronized and asynchronous beating of sailor cercariae in the wild. Note that the recordings were performed with the unique single individual ($n = 1$).
See also [Video S1](#).

formaldehyde solution for longer than 2 years and were tiny (with less than 1 mm^3 of biological material, largely anuclear), we opted for the approach previously used to isolate and sequence ancient DNA.⁶ The sailor sample yielded 72 contigs of more than 500 bp matching eukaryotic genomes, 36 of which matched Trematoda class (there were 46 unmatched contigs). The passenger sample yielded 359 contigs of more than 500 bp matching eukaryotic genomes, 42 of which matched Trematoda class (there were 132 unmatched contigs). We also found 28 Trematoda contigs present in both samples (more than 99% local BLAST similarity, [Table S2](#)). These trematoda-matched contigs contained little information about the features they contained, as the description of the BLAST hit from the closest match typically referred to an unannotated region of an assembled genome of the given species. Further analyses showed the presence of trematode ribosomal genes by blasting assembled contigs to SILVA small subunit and large subunit databases.⁷ HMMER algorithm 3.1⁸ predicted the presence of complete 18S, 5.8S, and 28S rRNA genes on two contigs, respectively coming

cercaria contains a yellow pigment in the hind part of the tail, which altogether creates a specific pattern across the entire aggregate ([Figures 2E–2G](#)). At the same time, the brown pigment of the passenger cercariae survived the fixation ([Figures 1C and 1D](#)). Morphological analysis revealed that this pigment is concentrated in numerous granules located in both the body and tail, mostly right under the tegument in a network-like way ([Figures 3B and 3F](#)). These granules appeared highly autofluorescent ([Figures 3G and 3H](#)). The results of electron microscopy validated the presence of these pigmented granules ([Figures 3I and 3J](#)), although their nature and functional meaning remained unknown.

Phylogenetic position

To understand the phylogenetic position of two morphotypes of cercariae, we separately sequenced fragments of their genomes including rRNA. Because the samples were fixed in a 10%

from the passenger and sailor samples (with a mean read coverage of 32.87). These genes from passengers and sailors were identical in sequence ([Data S1](#)) and aligned with known trematode 18S, 5.8S, and 28S rRNA genes from the Acanthocolpidae and Brachycladiidae families ([Figure 4A](#)). Three newly defined rRNA gene sequences from the sailor sample were submitted to GenBank under accession numbers OQ401338 (18S), OQ401339 (5.8S), OQ401340 (28S). The full-length rRNA-containing contig from the passenger sample was also submitted to GenBank under accession number OR242231.

For analyzed 28S and 18S rRNA gene sequences, the closest BLAST hits were species of the genus *Pleorchis*, family Acanthocolpidae. To resolve the phylogenetic position of our specimen, we used fragments of the 28S rRNA gene from 27 species of the Acanthocolpidae and their closest relatives Brachycladiidae ([Table S3](#)), which together form a monophyletic group with incompletely resolved interrelationships.^{9,10}

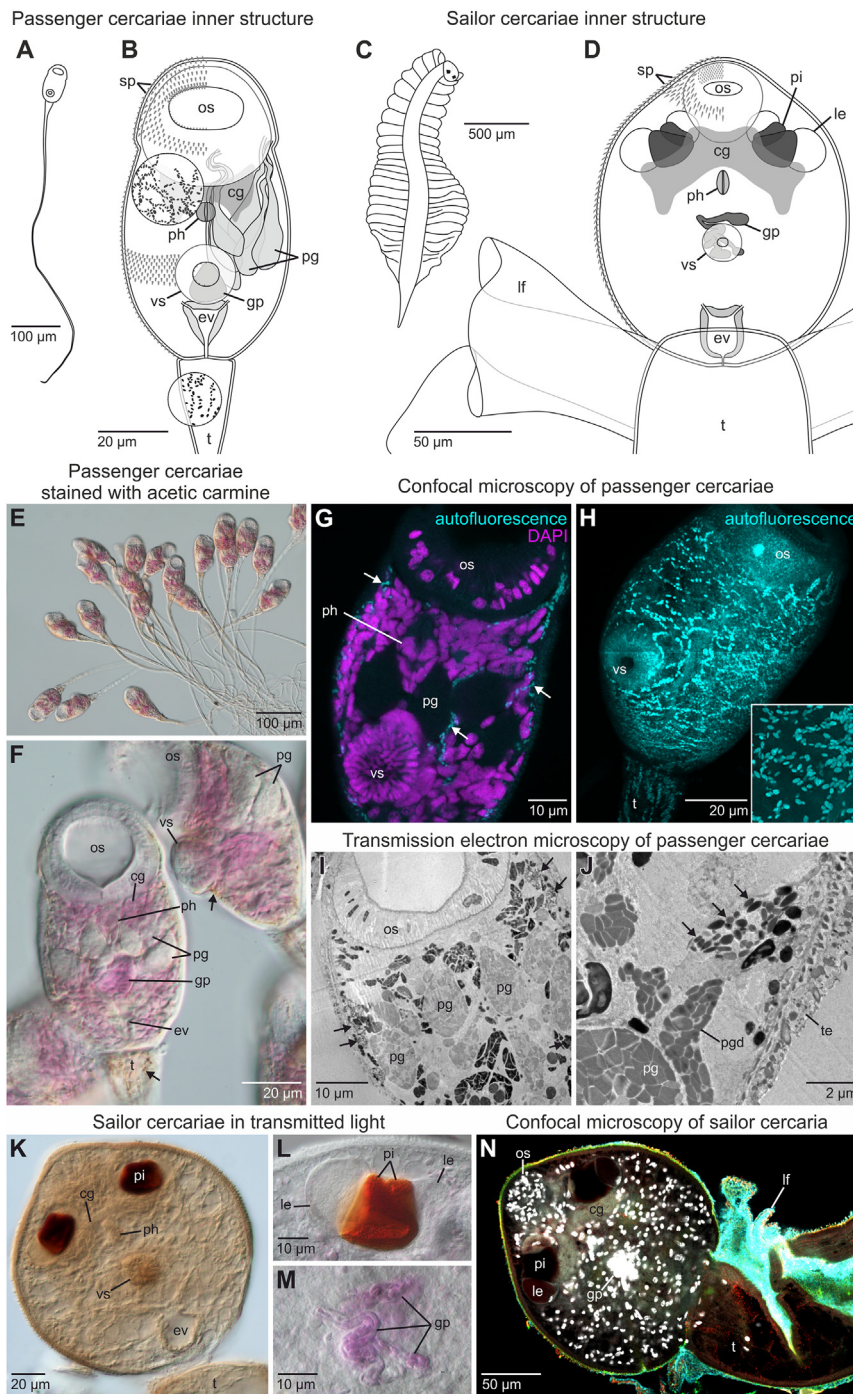


Figure 3. Inner structure of cercariae

(A–D) Schemes of passenger (A and B) and sailor (C and D) cercariae; distribution of pigmented granules is shown in circles.

(E and F) Whole mounts of passenger cercaria stained with acetic carmine.

(G and H) Confocal section of passenger cercaria (autofluorescence and DAPI).

(I and J) transmission electron microscopy (TEM) images of passenger cercaria.

(K) Body of sailor cercaria in differential interference contrast (DIC) light stained with Heidenhain's iron hematoxylin.

(L and M) Inner structure of sailor cercariae stained with acetic carmine.

(N) Confocal z stack based on autofluorescence and DAPI staining of sailor cercaria. Abbreviations: cg, cerebral ganglion; ev, excretory vesicle; gp, genital primordium; le, lens; lf, lateral fin of tail; os, oral sucker; pg, penetration glands; pgd, penetration gland ducts; ph, pharynx; pi, pigment cell; sp, spines; t, tail; te, tegument; vs, ventral sucker; arrows point at pigmented granules in passenger cercariae.

See also Figure S1 and Table S1.

of the Acanthocolpidae and Brachycladiidae, and *Paragonimus* spp. as the outgroup. In the obtained tree, our sequence formed a well-supported branch with *P. uku* within Acanthocolpidae clade A, though differed from it in 15 positions (Document S1).

Thus, the cercariae described here probably belong to the genus *Pleorchis* and are referred to from now on as *Pleorchis* sp. Okw1 ("Okw" for Okinawa).

DISCUSSION

The enigmatic swimming aggregations found in plankton near Okinawa Island appeared to be the colonies of the larval stages of the Digenea, cercariae. They probably represent a case of prey mimicry (prey mimetism), when the shape and behavior are aimed to attract the host and facilitate transmission. Locomotion type of the cercariae aggregate seems to copy movements of small-sized

Paragonimus spp. was added as the outgroup. The alignment was 1,367 bp long. Both maximum likelihood (ML) and Bayesian inference (BI) analyses resolved three monophyletic groups, similar to previous studies: Acanthocolpidae clades A and B and Brachycladiidae (Figure 4B). Possible paraphyly of the Acanthocolpidae had moderate support. Our specimen fell into the Acanthocolpidae clade A, into the genus *Pleorchis* with high support. It was closest to *P. uku* (10 substitutions).

We also ran ML and BI analyses for the 18S rRNA gene data. The alignment was 1,878 bp and included 23 sequences

plankton polychaetes, which undulate their bodies with high frequency while swimming. Prey mimicry is known from a variety of digenean taxa and also for bivalves from the order Unionida, which have parasitic larvae.^{3,4,11} One strategy in the digeneans is to have an extraordinary large tail imitating different animals, and it is exploited by the members of families Azygiidae, Bivesiculidae, Acanthocolpidae, and some others.^{9,12–15} Another strategy (and that is the present case) is the formation of a swimming aggregate of many cercariae adjoined together by their tails.⁴ Such aggregations are called "Rattenkönig" (Rat-king), and

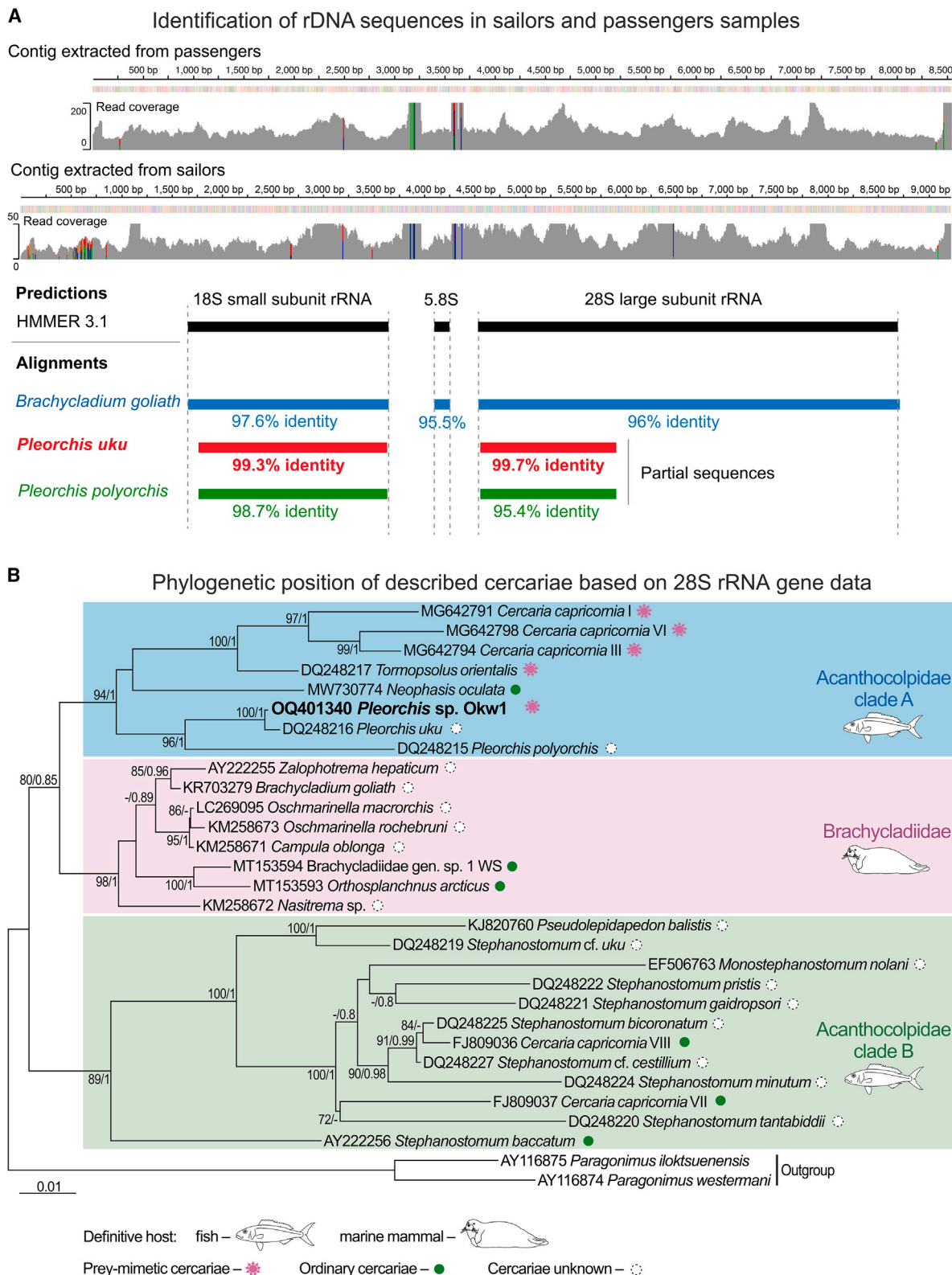


Figure 4. Analysis of ribosomal sequences

(A) Predictions of rRNA genes on two extracted contigs from sailors and passenger cercariae. On top, the two contigs are shown with gene coverage and aligned together by BLAST. In the center, horizontal black lines are predictions of rRNA sequences by HMMER. On the bottom, colorful horizontal lines are BLAST results from close species.

(legend continued on next page)

cercariae forming them are zygocercous. Zygocercous cercariae are known from various taxa: Heterophyidae,⁴ Cryptogonimidae,¹⁶ Diplostomoidea,¹⁷ Psilostomidae,¹⁸ and Renicolidae.¹⁹ Apparently, they evolved independently in these groups.⁴ Zygocercous cercariae fulfill three major goals: (1) prey-like appearance facilitates trophic transmission; (2) increased number of larvae infects the new host simultaneously; and (3) size of an aggregate helps the parasite to pass the host compatibility filter, as smaller fish cannot eat relatively large aggregate.^{3,4,17} However, so far, the zygocercous cercariae were known to make aggregates consisting only of one morphological type of larvae.

The peculiar feature of the discovered aggregation is the polymorphism of cercariae within it. Two morphs of cercariae, sailors and passengers, belong to one and the same species according to our molecular data, but they differ in structure, size, and behavior. They are also arranged in a specific way, but as the specimen is unique, we can't estimate variation in size and composition of such colonies.

Polymorphism seems to be quite rare in parasitic flatworms. In digeneans, it is mostly linked with dioecy of the sexual adults that occurs in the families Schistosomatidae and Didymozoidae.^{20,21} Polymorphism had also been proposed to be present in the asexual generations of the Digenea inhabiting the first intermediate hosts. Soldier and reproductive castes of rediae (parthenogenetic digenean stage) had been described, with the former assumed to serve for colony defense and to be unable to reproduce.^{22,23} However, other researchers suggest that these morphs are actually different stages of maturity and that soldiers do have reproductive organs.²⁴ Recently, somatic polymorphism was shown to occur in the free-living larvae of the Digenea: two types of cercariae different in size were described for *Trichobilharzia* sp. "peregrina," family Schistosomatidae.²⁵ These two morphs are assumed to diversify transmission strategies, but the general structure of these two morphs is quite similar. Thus, the aggregation-forming cercariae we described here demonstrate a first case of deep somatic polymorphism in digenean larvae.

To speculate on an adaptive value of such polymorphic colonies, a perspective from the life cycle is needed. According to the molecular data, the discovered cercariae belong to the clade within the putatively paraphyletic family Acanthocolpidae. They probably fall into the genus *Pleorchis* and thus got a temporary name *Pleorchis* sp. Okw1. Marine fish serve as the definitive hosts for *Pleorchis* spp. as well as for other acanthocolpids.²⁶ Life cycles for any of *Pleorchis* species are unknown but may be hypothesized to follow the basic scheme of the Acanthocolpidae: gastropod as the first intermediate host and fish as the second intermediate and definitive hosts.²⁷ Within the Acanthocolpidae clade A containing *Pleorchis*, such a life cycle had been described for *Neophasis oculata* and *Tormopsolus* spp.^{10,28} Zygocercous cercariae of *Pleorchis* sp. Okw1 probably aim to infect a fish second intermediate host: a fish tries to eat an aggregation, and the tail mass is ingested, but the bodies detach and

invade tissues of the buccal cavity, gills, etc.²⁹ The passenger cercariae of *Pleorchis* sp. Okw1 likely follow this strategy as they have well-developed penetration glands.

However, sailors lack penetration organs, and their destiny is questionable. Two options can be proposed. One is that sailors do not develop after a fish eats an aggregate; they sacrifice themselves to provide the transmission of passengers. This scenario would imply a strong division of labor between sailors and passengers. But the presence of genital primordium in sailors might be an argument against this hypothesis. Another option is that sailors continue their life in the fish that has eaten an aggregate. As penetration organs are absent, cercariae may go directly to the digestive system and develop into a sexual adult worm right there. In this case, the fish acts as the definitive host. A similar transmission route is known for cercariae of the Azygiidae and Bivesiculidae, which do not form aggregations but are also prey mimetic.^{12–14} Additional evidence for the latter hypothesis can be acquired from the information on feeding habits of the definitive hosts of *Pleorchis* spp. (Table S4). The majority of them are piscivorous, and that could provide transmission through the fish second intermediate host. But other documented definitive hosts of *Pleorchis* spp. feed on zooplankton (like *Caesio cuning* parasitized by *P. uku*), and in this case infection may be performed directly by cercariae.

Some members of the Acanthocolpidae clade A, namely *Tormopsolus* spp. and *Cercaria capricornia* groups 1–3, also have prey-mimetic cercariae.^{9,28} Moreover, there are several putative acanthocolpid prey-mimetic cercariae for which adults are unknown and molecular data absent.^{16,30} All these cercariae clearly differ in morphology from both sailors and passengers of *Pleorchis* sp. Okw1 and do not form aggregations; they have expanded hindbody or tail basement, and some of them resemble appendicularia (*Cercaria caribbea* LXXII) while others resemble zoea larvae (*Cercaria caribbea* LXXIII) or planktonic copepods or cladocerans (*Cercaria capricornia*). At the same time, *Neophasis oculata* from the Acanthocolpidae clade A has ordinary cercariae.¹⁰ Thus, it is highly probable that prey mimicry evolved in acanthocolpids independently several times.

In conclusion, the mystery surrounding the Okinawa sample has presented us with a new and fascinating example of a polymorphic colony of parasitic larval worms, which seem to exhibit prey mimicry. The physical integration, division of labor, and functional specialization of the two types of cercariae suggest a true colonial nature of the aggregate, comparable with known examples of specialized castes in insects.⁵ These newfound insights ignite a multitude of inquiries and potential avenues for future research, such as the examination of the developmental paths of the various cercariae types comprising the aggregate and the genes responsible for phenotypic shifting.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

(B) Phylogenetic position of described cercariae resulting from maximum likelihood (ML) and Bayesian inference (BI) analyses based on 28S rRNA gene sequence data. Bootstrap values are given in nodes, followed by posterior probabilities. Support values lower than 70 (ML) and 0.80 (BI) are not shown. Scale bar shows number of substitutions per site. Newly generated sequence is in bold.

See also Figure S2, Tables S2, S3, S4, and Data S1.

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2023.08.090>.

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AUTHOR CONTRIBUTIONS

D.K., A.M., E.K., L.F., R.M., L.H., M.L., P.K., M.E.K., and I.A. performed experimental work, analyzed the results, and participated in drafting the manuscript. I.A. conceived the study and secured the funding.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-GAD	Sigma-Aldrich	Cat#ABN904; RRID:AB_2893025
Alexa Fluor® 488 Phalloidin	Cell Signaling	Cat#8878
Donkey anti-Rabbit IgG secondary antibody, Alexa Fluor™ 488	Thermo Fisher Scientific	Cat#A-21206; RRID:AB_2535792
Donkey anti-Rabbit IgG secondary antibody, Alexa Fluor™ 647	Thermo Fisher Scientific	Cat#A-31573; RRID:AB_2536183
Donkey anti-Rabbit IgG secondary antibody, Alexa Fluor™ 555	Thermo Fisher Scientific	Cat#A-31572; RRID:AB_162543
Biological samples		
Fixed specimens of <i>Pleorchis</i> sp. Okw1 cercariae	Natural History Museum Vienna	Evertebrata varia collection: NHMW-ZOO-EV-M-5885
Chemicals, peptides, and recombinant proteins		
"Eukitt(R)" quick hardening mounting medium	Sigma-Aldrich	Cat#03989
Bio Mount medium	Bio Optica	Cat#05-BMHM508
Carmine	Sigma-Aldrich	Cat#C1022
Hematoxylin	Sigma-Aldrich	Cat#H9627
Phosphate buffered saline	Sigma-Aldrich	Cat#P4417
DAPI	Sigma-Aldrich	Cat#D9542
Tween-20	Sigma-Aldrich	Cat#11332465001
LX 112 resin	Ladd	Cat#21310
GlycoBlue	Ambion	Cat#AM9516
Critical commercial assays		
SMARTer ThruPLEX DNA-seq library preparation kit	Takara Bio	Cat#R400674
NovaSeq 6000 S4 Reagent Kit v1.5	Illumina	Cat#20028312
Deposited data		
Raw sequencing reads	SRA	BioProject number PRJNA931659
<i>Pleorchis</i> sp. Okw1 sailor cercariae 18S rRNA gene, partial sequence	GenBank	Accession number: OQ401338
<i>Pleorchis</i> sp. Okw1 sailor cercariae 5.8S rRNA gene, partial sequence	GenBank	Accession number: OQ401339
<i>Pleorchis</i> sp. Okw1 sailor cercariae 28S rRNA gene, partial sequence	GenBank	Accession number: OQ401340
<i>Pleorchis</i> sp. Okw1 passenger cercariae 18S rRNA gene, partial sequence; ITS1, 5.8S rRNA gene ITS2, complete sequences; and 28S rRNA gene, partial sequence	GenBank	Accession number: OR242231
Software and algorithms		
Fiji (ImageJ) software	Schindelin et al. ³¹	https://fiji.sc/
Imaris software	Oxford instruments	https://imaris.oxinst.com/
Snakemake ³² pipeline	GitHub	https://github.com/LouisFaure/pleorchis_project
Trimmomatic software	Bolger et al. ³³	https://academic.oup.com/bioinformatics/article/30/15/2114/2390096
SPAdes computational tool	Bankevich et al. ³⁴	https://cab.spbu.ru/software/spades/
BWA-MEM computational tool	Li ³⁵	https://github.com/lh3/bwa
HMMER 3.1 algorithm	Wheeler, Eddy ⁸	https://academic.oup.com/bioinformatics/article/29/19/2487/186765
barnap 0.9 computational tool	Seemann ³⁶	https://github.com/tseemann/barnap

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
MEGA11 computational tool	Tamura et al. ³⁷	https://www.megasoftware.net/
IQ-TREE 1.6.12 computational tool	Nguyen et al. ³⁸	http://iqtree.cibiv.univie.ac.at/
UFBoot2 computational tool	Hoang et al. ³⁹	N/A
MrBayes v.3.2.7a computational tool	Ronquist et al. ⁴⁰	http://nbisweden.github.io/MrBayes/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Igor Adameyko (igor.adameyko@ki.se).

Materials availability

Fixed specimens were deposited to The Natural History Museum Vienna, please refer to the Evertabrata varia collection: NHMW-ZOO-EV-M-5885.

Data and code availability

- Raw sequencing reads have been deposited to SRA under BioProject number PRJNA931659 and are publicly available as of the date of publication. The newly identified sequences of 18S, 5.8S and 28S rRNA genes from sailor cercariae and the full length rRNA-containing contig from passenger cercariae are deposited in GenBank and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#).
- The computational analysis of the sequencing data is made available as a completely reproducible snakemake³² pipeline on the following GitHub repository: https://github.com/LouisFaure/pleorchis_project
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

A single aggregate of *Pleorchis* sp. Okw1 cercariae was collected on 11th of April 2018 at 16m depth in Okinawa waters (25°.990152"N, 127°.611343"E; off shore of Kiyon Cape Itoman-city Okinawa, Japan; water temperature 24 °C). It was filmed swimming in a laboratory glass in seawater, and then immediately fixed for further studies.

METHOD DETAILS

Fixation

The specimen was fixed in a 10% non-buffered formaldehyde solution and kept in the same fixative at room temperature for two years. Pieces of specimen were then transferred either in 96% ethanol (for histological staining, transmission and scanning electron microscopy, and DNA extraction), or 70% methanol (for immunohistochemistry) and kept at 4°C from that point till the analysis stage, which took between 1 and 12 months for different portions of the sample.

An unstained piece of specimen for the museum slide was mounted with the "Eukitt(R)" quick hardening mounting medium (Sigma-Aldrich, #03989).

Histological staining of whole mounts

For whole mounts, specimens were rinsed in distilled water, and stained either with acetic carmine followed by 0.1 M HCl in 70% ethanol destaining, or with Heidenhain's iron hematoxylin followed by picric acid destaining. After, specimens were dehydrated in graded ethanol series, cleared in xylol, and mounted in Bio Mount medium (Bio Optica, Italy). Slides were examined under a Leica DM2500 microscope (Leica Microsystems, Germany) with DIC, and photos were taken with a Nikon DS Fi3 (Nikon, Japan) camera and processed with Fiji software.³¹ Staining by Heidenhain's hematoxylin wasn't successful, it gave uniform light brown color instead of dyeing nuclei and muscle fibers in black (probably due to fixation); however, obtained mounts were appropriate to visualize general structure of cercariae and to take measurements. All cercariae used for the descriptions come from a single aggregate. All measurements are in micrometers.

Immunohistochemistry and confocal microscopy

Whole samples were brought to room temperature in the glass jars and washed twice with PBS after 70% methanol. Then they were transferred to new jars by pipetting under the stereomicroscope. Then the samples were incubated overnight on a rocker at room

temperature with primary antibodies and DAPI (4',6-diamidino-2-phenylindole) (5 µg/mL) diluted in PBS containing 0.1% Tween-20 (PBST). After incubation with primary antibodies, the slides were washed 3 times in PBST (10 min every wash) on a rocking platform at a room temperature. Then, the secondary Alexa antibodies were added to the slides in PBST and incubated for 2 h at room temperature. After the incubation with secondary antibodies, the slides were subjected to 3 sequential washes in PBST (room temperature, rocking platform) and mounted with glycerol for confocal microscopy.

The following primary antibodies were used: rabbit anti-GAD (1:500, Sigma-Aldrich, ABN904), animal anti-F-actin (anti-Phalloidin) (1:1000, Cell Signaling, #8878). For detection of the primary antibodies, secondary antibodies raised in donkey and conjugated with Alexa 488, -555, and -647 fluorophores were used (1:1000, Molecular Probes, ThermoFisher Scientific).

Images were acquired using LSM 880 confocal microscopes (Carl Zeiss, Germany) with ×20, ×40, ×63, ×100 objectives and AiryScan module. Tile and z stack functions were used for obtaining the images, AiryScan function was performed for the images obtained with ×40, ×63 and ×100 objectives. Images were acquired in the .lsm format, 3D reconstruction was performed in Imaris (Imaris Converter, Imaris).

Transmission electron microscopy (TEM)

The cercariae specimens from 96% ethanol were rinsed in 0.1M phosphate buffer prior to postfixation in 2% osmium tetroxide in 0.1M phosphate buffer, pH 7.4 at 4°C for 2 h. Following stepwise dehydration in ethanol and acetone, the specimens were embedded in LX-112 resin (Ladd). Ultrathin sections (approximately 80–100 nm) prepared using a EM UC7 (Leica) and contrasted with uranyl acetate followed by lead citrate were examined in a HT7700 transmission electron microscope (Hitachi High-Technologies,) at 80 kV. Digital images were acquired using a 2kx2k Veleta CCD camera (Olympus Soft Imaging Solutions).

Scanning electron microscopy (SEM)

The specimens were adhered onto PES pore membranes and washed with MilliQ water prior to stepwise ethanol dehydration and critical-point-drying in an EM CPD300 (Leica) or immersed in tetramethylsilane and allowed to air dry at room temperature. The pore membranes were mounted on aluminum specimen pins using carbon adhesive tabs and sputter coated with platinum using a Q150T ES (Quorum). SEM images were acquired using an Ultra 55 field emission scanning electron microscope (Zeiss) at 3kV and the SE2 detector.

DNA extraction

DNA extraction was performed for the large (sailors) and small (passengers) cercariae separately, using a published protocol.⁶ 0.5 mL of an alkaline digestion buffer (0.1 M NaOH with 1% SDS solution) was added to 2-mL screw-cap O-ring tubes containing tissue. The tubes were placed in an autoclave and heated to 120°C for 25 min. Following the heating, the tubes were left at room temperature for 5 min. 500 µL of a 25:24:1 phenol:chloroform:isoamyl alcohol solution was added to each tube which were gently agitated at room temperature for 5 min. The tubes were then centrifuged for 5 min at 15000 × g to separate the upper aqueous layer from the lower phenol layer. The upper aqueous layer was moved to a new tube containing 500 µL chloroform, which was followed once again by 5 min of centrifugation at 15000 × g. The upper aqueous layer was moved to a new 1.5 mL Eppendorf tube. After estimating the volume of the extracted layer, 0.6 volume isopropanol and 0.1 volume 3 M sodium acetate (approx. pH 5) were added to the tube. A commercial carrier solution, Glycoblue (Ambion, Inc., Austin, TX), was added to facilitate pellet visualization. The tube was mixed and then centrifuged at high speed (15000 × g) for 30 min at room temperature. Following centrifugation, the liquid was decanted from the tube, and the DNA pellet was rinsed with 500 µL 85% ethanol and centrifuged for 5 min at high speed (15000 × g). The ethanol was then decanted, and any residual ethanol was removed from the pellet, which was then briefly incubated at 65°C for 30s. The pellet was re-suspended in 60 µL of ddH₂O.

DNA sequencing

Extracted DNA from the large (sailors) and the small (passengers) cercariae were sent to SNP&SEQ Platform for library preparation and sequencing. Library preparation was performed using the SMARTer ThruPLEX DNA-seq library preparation kit (Takara Bio). One library per sample was prepared. The 2 libraries were separately loaded into each lane of the Illumina SP flow cell according to Xp workflow, allowing us to obtain distinct datasets for each library and minimize any potential cross-contamination or bias. Paired-end sequencing of the libraries was performed over 150 cycles using a SP flowcell with the NovaSeq 6000 system and v1.5 sequencing chemistry (Illumina Inc.). The sequenced reads were subsequently provided in gzip-compressed fastq-format. The total reads from contigs longer than 500bp amounts to 6,740,347 for the sailors sample, and to 15,250,636 for the passengers sample. The difference is explained by the fact that we used the lower quality undetermined reads for the latter which contains many reads from index adapters merged with the contigs. Reads related to contigs matching Trematoda class amounts to 2,700,315 for the sailors sample and 2,119,045 for the passengers sample.

QUANTIFICATION AND STATISTICAL ANALYSIS

Computational analysis of DNA sequences

Computational analysis was performed on a 88 CPU cores and 1Tb RAM server. Analysis was performed first on both DNA samples, which contained high quality demultiplexed data. Although passengers demultiplexing failed, data could be recovered by using

“undetermined” reads from the flow cell lane where passengers libraries were loaded. In all cases, fastq files of forward and reverse reads were trimmed using trimmomatic³³ with a sliding window of 5:20 and a minimum length of 50. The trimmed paired reads were error corrected using SPAdes tool.³⁴ For the passengers sample, an additional run of trimmomatic was performed to remove specific illumina index sequences that would affect the assembly (AATGATACGGCGACCACCGA and its reverse complement: TCGGTGGTCGCCGTATCATT). Assembly was performed with the SPAdes on the error corrected forward and reverse reads, leading to contigs. Sequences shorter than 500 bp were discarded. The extracted contigs have been uploaded on figshare (10.6084/m9.figshare.23946969). A BLAST nucleotide search was performed on these contigs using custom built databases from SILVA (release 138.1)⁷ for both SSU and LSU sequences. Only sequences matching the Holozoa clade were kept. In each sailors and passenger samples, a single contig was found to match SSU and LSU from Trematoda class. For each SSU/LSU-containing contig found, reads from the related sample were processed using bwa mem³⁵ to obtain coverage and mismatch information. 18S, 5.8S and 28S ribosomal RNA genes were predicted by applying HMMER 3.1 algorithm⁸ on each contig (using barnap wrapper³⁶). The predicted rRNA genes were subsequently confirmed by homology with the 18S, 5.8S and 28S rRNA genes from trematode *Brachycladium goliath* (GenBank KR703279.1), using local BLAST alignment. Finally, the single contig was blasted locally to a set of two partial sequences of 28S rRNA gene from *Pleorchis uku* (GenBank DQ248216.1) and *Pleorchis polyorchis* (GenBank DQ248215.1).

As a part of the analysis, we identified and retained contigs that matched Trematoda and were present in both samples. Overlap of contigs between the two samples were performed as following: each trematoda contigs from passengers sample were BLASTed locally against all trematoda contigs from sailors. After conducting a BLAST search against the whole nucleotide database, we found that most of the contigs are unidentified (except for some hits on the mitochondrial genome), as the description of the BLAST hit from the closest match typically refers to an unannotated region of an assembled genome of the given species. Overall, there are no fully annotated genomes close enough to our species that would allow us to identify genes or features that are not ribosomal genes or part of the mitochondrial genome. We have included Table S2, showing the name of the overlapping contigs from both samples, with the description of the closest Trematoda hit.

In our study, “unmatched” refers to contigs that either did not have a hit in the BLAST against the NCBI nucleotide database, or were not classified within the known taxonomies (artificial, Viruses, Bacteria, Eukaryota, Trematoda). In the latter case, this does not necessarily mean that they do not match anything in the databases, but rather that they did not match any of the categories we were specifically looking for. Unfortunately, there is currently no specific machine learning tool that can directly input completely unmatched genomic sequences and output a prediction of whether the sequence is eukaryotic or prokaryotic or virus.

The higher number of unmatched sequences in passengers could be due to the lower quality of these reads, as we used ancient DNA sequencing method. We also took in the account the undetermined reads, which often contained errors or ambiguities in the index sequences used for multiplexing. Despite our efforts to trim a specific Illumina index from the raw reads to reduce the number of chimeras, it’s likely that not all were removed due to sequencing errors. Thus, these unmatched contigs could potentially match other categories not included in our taxonomy classification or could be a result of read errors.

Phylogenetic reconstructions

Taxa selection for the phylogenetic analysis was based on BLAST results: as the closest hits to our sequences were members of the genus *Pleorchis* (family Acanthocolpidae), we took from GenBank all available sequences of 28S and 18S rDNA of the Acanthocolpidae and their closest relatives Brachycladiidae. After provisional alignments were built, several much shorter sequences were removed from the analysis. The family Paragonimidae (two species, *Paragonimus iloktsuenensis* and *P. westermanni*) was selected as an outgroup as it appeared the closest relative of Acanthocolpidae+Brachycladiidae in recent phylogenetic reconstructions.⁴¹ Data from GenBank included in 28S and 18S rRNA gene alignments are listed in Table S3. The best model was selected via MEGA11,³⁷ it was GTR+G+I for 28S dataset and K2+G+I for 18S dataset. ML analysis was run in IQ-TREE 1.6.12³⁸ and UFBoot2 (ultrafast bootstrap³⁹) with 1000 replicates. The BI analysis was conducted in MrBayes v.3.2.7a⁴⁰ with 10,000,000 generations.